Assignment 1: Part A
Question Formation and Exploratory Analysis
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Introduction

Initial Question:

Are females with a specific blood group more prone to developing Polycystic Ovary Syndrome (PCOS)?

PCOS, a common hormonal disorder among women of childbearing age, causes hormonal imbalances leading to irregular menstrual cycle, heightened androgen[1] levels, male-pattern hair loss, acne[2], infertility, excess hair growth, and abdominal weight gain (World Health Organization 2023). While symptoms can be managed through diet, lifestyle adjustments, and medication, there's no cure (World Health Organization 2023). PCOS also contributes to psychological challenges like anxiety, depression, and poor body image, impacting holistic existence (World Health Organization 2023).

PCOS is a complex disorder with an uncertain origin. This study aims to investigate the potential correlation between blood groups and PCOS. Insights from this research could enhance our understanding of the underlying mechanisms and risk factors of PCOS. If we find a specific blood type that's more likely to have PCOS, we can do early checks and treatments for those people. This could help personalize medicine and improve health outcomes for women. It could also lead to better public health plans and education to help prevent PCOS or reduce its severity.

[1] Androgen: a group of hormones that play a role in male traits and reproductive activity.
[2] Acne: a common inflammatory skin condition caused by clogged hair follicles connected to oil glands.

Data Information

Datasets were collected from Kaggle, a trusted site known for good quality data, ensuring that the data is credible in the research community (www.kaggle.com n.d.). After collating these datasets, the final dataset includes the following columns:

Table 1: Variables present in the dataset and their description.

Type variable	**Column Names**	**Description**
Categorical with binary values.	PCOS	Indicates if the individual has PCOS (Yes/No).
	Regular menstrual cycle	Indicates if the individual has regular menstrual cycle (Yes/No).
	Weight gain	Indicates whether the individual has had weight gain (Yes/No)
	Excessive body/facial hair	Indicates the presence of hirsutism (Yes/No).
	Skin darkening	Indicates presence of skin darkening in any part of the body (Yes/No).
	Hair loss/baldness	Indicates whether the individual has experienced hair loss/baldness (Yes/No).
	Pimples on face/jawline	Indicates whether the individual experiences acne (Yes/No).
	Regular fast food intake	Indicates if the individual regularly consumes fast food (Yes/No).
	Regular physical activity	Indicates if the individual does any regular physical activity (Yes/No).
Quantitative discrete	Age (yrs)	The age of the individual.
	Cycle length (days)	Mentions the number of days the individual has her each menstruation.
Quantitative continous	Weight (Kg)	The weight of the individual.
	Height (cm)	The height of the individual.
Categorical	BMI	Body Mass Index, calculated from weight and height.
	Blood Group	The blood group of the individual.

The mutilple datasets, the variety and the complexity of the variables aligns with the definition of big data.

Data Processing

For data collection, four distinct datasets were collected from Kaggle. Missing values were identified in several rows for blood group, weight, height, and BMI in two datasets.To address the missing blood group values, the overall distribution of each blood group was calculated, and the missing values were then randomly imputed in the same proportion. BMI values were calculated using the available weight and height data for the respective rows. For the missing weight and height values, the mean weight and height for each BMI category were determined and assigned to the rows with missing data. Duplicate rows were removed and outliers for the quantitative data was checked and those rows were removed.

The data was analysed, the categorical variables with binary values were encoded as Yes/No for consistency from 0/1 or other numerical indicators. Numerical BMI values were converted into standard BMI categories based on ranges from betterhealth.vic.gov.au. The datasets then were integrated, standardizing the column names across all the datasets, and removing unrequired columns to ensure consistency.

Finally, the imputed values were verified for discrepancies, ensuring the dataset's relaibility for subsequent analysis.

The dataset now includes key variables related to PCOS symptoms and blood group. Factors like cycle regularity, weight gain, and lifestyle habits (fast food intake, physical activity) give a complete view of possible interactions. It also covers important demographic (age) and physical (weight, height, BMI) details to control for other influences.

Limitations

- **Missing data:**
  There were few missing data which were handled using imputation technique.

- **Data duplication:**
  Duplicate records were carefully identified and removed to prevent potential bias.

- **Bias in data:**
  The data might be biased, potentially limiting its generalizability. Certain ethnic groups or age ranges might have been underrepresented. This bias cannot be entirely eliminated. However, future studies could aim to collect data from a more diverse population to reduce bias.

- **Limited sample size:**
  The sample size isn't large enough to detect small effects. Data isn't available easily as this is a sensitive information regarding women's reproductive health. More samples can be obtained from conducting surveys or seeking for help from research databases.

Refined question: Are females of age 15-45 years with a specific blood group more prone to developing Polycystic Ovary Syndrome (PCOS)?

After data processing, the next steps include exploratory data analysis and calculating descriptive statistics for quantitative variables. Then, visualizing the data using plots for the variables. Identifying correlations between the variables, testing for significant differences, and then fitting a model using logistic regression and evaluating it using metrics such as accuracy, precision, recall, and ROC curve.

Backup question: How does PCOS affect fertility across different age groups?

To work on this question, along with the existing dataset information like number of pregnancies, infertility diagnosis, hormonal profiles like LH[3],FSH[4],AMH[5], and information of ovarian cysts through ultrasound images will be required (Mayo Clinic 2021).

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[3] LH: Luteinising hormone, crucial in regulating the function of ovaries in women.
[4] FSH:Follicle Stimulating Hormone that helps in the release of an egg each month (ovulation) in females
[5] AMH: Anti-Mullerian Hormone, a popular hormone that is typically used to examine ovarian reserve and response to fertility treatments.

References

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